SMARTBRIDGE
Let's Bridge the Gap

SKILL WALLET
A SmartBridge Product

Project Executable Files

Date:	29 june2026
Team ID:	*(xxxx)*
Project Name: Learning	Human Development Index (HDI) Prediction Using Machine
Maximum Marks:	3 Marks

Project Executable Files The **Human Development Index (HDI) Prediction System** is developed using **Python, Flask, Scikit-learn, Pandas, NumPy, HTML, CSS, JavaScript, and Visual Studio**
Code. The project predicts the HDI category of a country based on **Life Expectancy, Mean Years of Schooling, Expected Years of Schooling, and GNI Per Capita**. All required source code, model files, datasets, and documentation are organized for easy execution and evaluation.

Step 1: Submission Checklist:_

S.No	Item to Submit	Submitted
1	Complete source code (all files and folders)	**Yes**
2	README / Setup Guide	**Yes**
3	requirements.txt (dependency file)	**Yes**
4	Database schema / seed files (Not Applicable – CSV dataset used)	**NA**
5	Environment configuration file (.env.example)	**NA**
6	Deployed application URL	**No** *(Add if deployed)*
7	Executable binary / APK	**NA**
8	Dockerfile / Containerization configuration	**NA**

9	Test files and test results	**Yes**
S.No	**Item to Submit**	**Submitted**
10	Demo video / Walkthrough	**Yes** *(If required by your mentor)*

Step 2: File / Folder Structure:-

```
HDI-Prediction/
|
├── dataset/
|   └── hdi_dataset.csv
|
├── models/
|   └── model.pkl
|
├── static/
|   ├── style.css
|   └── script.js
|
├── templates/
|   └── index.html
|
├── app.py
├── train_model.py
├── requirements.txt
├── README.md
├── test_prediction.py
```

└── .gitignore

Step 3: Deployment / Access Details

Field	**Details**
Hosted / Deployed URL	*Add your deployed application URL (if hosted on Render, Railway, or PythonAnywhere).*
Login Credentials (Demo)	**Not Required** (The application does not require login.)
Platform / Hosting Provider	Local Flask Server (Development). Future deployment can use **Render**, **Railway**, or **PythonAnywhere**.
Repository Link	*Paste your GitHub repository URL here.*
Demo Video Link	*Paste your demo video link here (Google Drive/YouTube, if required).*

Step 4: Run Instructions:- Pre-requisites

- Windows / Linux / macOS
- Python 3.x
- Visual Studio Code
- Flask
- Pandas
- NumPy
- Scikit-learn
- Joblib

Steps

1. Clone the repository:
2. git clone <GitHub Repository URL>
3. Open the project in **Visual Studio Code**.
4. Install all required dependencies:

5. pip install -r requirements.txt
6. Verify that hdi_dataset.csv and model.pkl are available in the appropriate folders.
7. Start the Flask application:
8. python app.py
9. Open a web browser and visit:
10. http://127.0.0.1:5000
11. Enter:

    ○ Life Expectancy ○ Mean
    Years of Schooling ○
    Expected Years of Schooling ○
    GNI Per Capita

12. Click the **Predict** button to view the predicted HDI category.

Step 5: Known Issues / Limitations:-

Known Issue / Limitation	**Workaround / Status**
Uses a local dataset only	Can be enhanced by integrating live UNDP data in future versions.
Local Flask server is intended for development	Deploy on a production server such as Render or Railway for wider access.
Prediction depends on the quality of the training dataset	Improve accuracy by expanding and updating the dataset regularly.